Hypothyroidism

(NICE 2019, NG145, BJGP 2016;66:538, BMJ 2008;337:a801)

Red Whale
GEMS
Guidelines & Evidence Made Simple

Diagnosis of hypothyroidism in adults (this GEMS is NOT applicable to pregnant women/trying to conceive)

Do not diagnose based on symptoms alone.
If suspected, NICE recommends checking TSH, and, if abnormal, check T4 (free thyroxine) (most labs add this).
If pituitary disease suspected (secondary thyroid dysfunction), check TSH and T4.
Free T3 testing is not recommended for diagnosis or monitoring of hypothyroidism.
Repeat tests if symptoms worsen/new symptoms develop; wait 6w from initial test before repeat.

Test if:	*Consider* testing if:
• Symptoms of hypothyroidism. • New-onset atrial fibrillation. • Type 1 diabetes/other autoimmune disorder.	• Depression. • Unexplained anxiety.

In children/young people, consider testing in all the above situations PLUS:
- Abnormal growth.
- Unexplained changes in behaviour/school performance.

A BMJ review also suggested annual TSH in those (BMJ 2008;337:a801):
- With Down's/Turner's syndrome.
- Taking specific drugs: lithium, amiodarone, rifampicin, thalidomide, interferons and sunitinib.
- Who have had radioiodine treatment/neck radiotherapy/subtotal thyroidectomy.

- Do not offer testing solely because someone has type 2 diabetes.
- Do not test thyroid function at the time of an acute illness (can affect results).

Interpreting blood results

TSH raised and T4 low: HYPOTHYROIDISM → Offer levothyroxine (below)

Consider checking thyroid peroxidase antibodies in those with raised TSH (high TPO levels suggest autoimmune cause)

TSH raised and T4 normal: SUBCLINICAL HYPOTHYROIDISM → See GEMS on SUBCLINICAL hypothyroidism

Offer levothyroxine for confirmed HYPOTHYROIDISM

- **Offer levothyroxine:**
  - Levothyroxine should be taken at least 30 minutes before breakfast, caffeine and other drugs (BNF).
  - If ≥65y or known cardiovascular disease, start at 25–50micrograms/day and titrate up as required.
  - For everyone else, straight to estimated treatment dose: 1.6micrograms/kg/day (to the nearest 25mcg) (for a 60kg person = 100mcg, for an 80kg person = 125mcg, for a 100kg person = 150mcg).
- Do NOT offer natural thyroid extract or liothyronine (see over for more information).

Commonest cause of hypothyroidism is autoimmune Hashimoto's thyroiditis.

Aim of treatment and monitoring

REMEMBER: after thyroid cancer, oncology often want long-term TSH suppression (<0.1); check if uncertain!

For everyone else, once on long-term levothyroxine:
- Aim for normal TSH. No benefit in aiming for a TSH at lower end of normal.
- If continued symptoms, can adjust dose to optimise symptoms but TSH should not be supressed *(increases the risk of osteoporosis and atrial fibrillation)*.
- If continued symptoms after starting levothyroxine, consider checking T4 as well as TSH.

Monitoring:
- Consider checking TSH every 3 months until stabilised (2 similar measurements within normal range 3m apart). TSH can take up to 6m to normalise if the initial TSH was very high/has been raised for a long time.
- **Once stable on treatment, check TSH annually.**

Hypothyroidism is associated with cardiovascular risk factors/increased risk of cardiovascular disease.

Troubleshooting in hypothyroidism

(NICE 2019, NG145, BMJ 2021;373:n993, BJGP 2016;66:538, BMJ 2008;337:a801, BMJ 2019;364:l579, NEJM 2019;381:749, MHRA Drug Safety Update May 2021, Clinical Endocrinology 2023;99:206)

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Diagnosis troubleshooting: symptoms of hypothyroidism but a normal TSH

- If there is pituitary pathology, T4 can fall in the presence of a normal TSH. This is rare.
  - Assess for symptoms/signs of pituitary disease, measure T4, consider measuring other pituitary function bloods.
- The use of biotin supplements (vitamin B7) can raise or lower TSH. If taking: stop and retest TSH several days later.

Management of clinical hypothyroidism troubleshooting (raised TSH, low T4)

Liothyronine

There is no long-term safety data for T3 (liothyronine)

NICE recommendation:
Liothyronine is not recommended alone or in combination (lack of evidence of efficacy), BUT, for some patients, liothyronine is used if there is a lack of symptomatic response to levothyroxine, and its use may benefit small subgroups of patients.

The Joint British Thyroid Association and Society for Endocrinology guidelines say:

- Most will achieve satisfactory and appropriate control with levothyroxine alone.
- Some on levothyroxine have ongoing symptoms despite appropriate biochemical correction.
- Endocrinology review is advised if considering liothyronine. Endocrinology may <u>trial</u> liothyronine if:
  - Good compliance with levothyroxine AND persistent symptoms despite normal TSH AND comorbidities excluded AND there is a confirmed diagnosis of overt hypothyroidism AND TSH has been maintained towards the lower end of reference range (e.g. 0.3–2.0 mU/L) for 6m.
- If considering stopping T3, this should be done under the supervision of an endocrinologist.

Still symptomatic but TSH is normal

- Are there other autoimmune conditions?
- Consider nutritional deficiencies, e.g. B12, folate, vitamin D.
- Consider other causes of TATT or hypothyroid-like symptoms:
  - Obesity, sleep apnoea, menopause, anxiety, depression, stress, lifestyle, drugs, alcohol.
  - Medications, e.g. beta-blockers, opioids, corticosteroids, benzodiazepines.

Treatment-refractory hypothyroidism

Suspect if: persistent symptoms OR raised TSH DESPITE optimum levothyroxine (up to 2mcg/kg body weight).

Look for these causes and then recheck TFTs 6w after any changes: consider referral to endocrine if TSH remains elevated.

Are they taking their medication? If not, is there a specific reason as to why?	• Review compliance. • Develop shared management plan, including patient's goals/preferences. • Discuss risks of non-adherence, including weight gain, tiredness, depression, raised cholesterol and myxoedema coma (which can be fatal).
Are they taking thyroxine with other drugs/meals which can affect absorption? • Food, coffee, soya, kelp, grapefruit juice. • Multivitamins, calcium, iron, antacids. • Drugs: PPIs, cholestyramine, colestipol, sevelamer, sucralfate, raloxifene. **Is there an increased need for thyroxine?** • Weight gain. • Pregnancy. • Cytochrome p450 inducers: phenytoin, carbamazepine, phenobarbital, rifampicin.	• Take thyroxine on an empty stomach with water 1h before breakfast/other tablets to maximise absorption. • A levothyroxine dose increase of 25–50micrograms/d may be appropriate for patients who need to take other medications that decrease its bioavailability.
Do they have a diagnosis of/any signs of intestinal malabsorption? • Coeliac disease, inflammatory bowel disease, lactose intolerance, short bowel syndrome, infiltrative enteropathy. • Infections, e.g. giardia, helicobacter pylori.	• Check bloods for malabsorption: FBC, B12, folate, ferritin, calcium, albumin and coeliac serology. • Refer as appropriate if a malabsorptive condition is suspected.

Symptoms with a different levothyroxine brand (MHRA)

Levothyroxine is usually prescribed generically. A small number of patients report symptoms if the brand is changed. If this occurs, consider testing thyroid function.

- If symptoms persist (whether thyroid function is normal or abnormal), consider consistently prescribing a tolerated brand.
- If symptoms/poor control persists, consider prescribing levothyroxine oral solution.

Drug interactions

- Drugs which can cause hypothyroidism: lithium (6 times increased risk) and amiodarone.
- Drugs affected by hypothyroidism:
  - Statins: increased risk of myopathy when used in patients with known hypothyroidism.
  - Propranolol, steroids and digoxin: clearance of drugs delayed.